Lakshya Jindal

+918169088532 lakshyajindal0310@gmail.com linkedin.com/in/lakshyaj03 GitHub Codeforces

Education

International Institute of Information Technology, Hyderabad Aug 2023 – Present
B.Tech + MS(By Research) in Electronics and Communication Engineering *Hyderabad, India*

Hiranandani Foundation School Thane June 2021 – June 2023
ISC, PCM and Computer Applications - 94.25% *Mumbai, India*

Experience

Undergraduate Researcher | Robotics Research Centre, IIIT Hyderabad April 2025 – Present
Research Assistant *On-site*

- Designed and implemented scalable algorithms in C++ and Python, applying a solid foundation in linear algebra, probability, and statistics to solve high-dimensional numerical optimization problems.
- Built modular software pipelines for large-scale algorithm evaluation using Linux and Git, prioritizing robust software engineering principles, test automation, and reproducibility.

Publications

Towards Kinematic Actionable Infeasibility Detection in Motion Planning | *IEEE RA-L (Accepted)*

- Co-authored research introducing a geometry-driven framework for certifying infeasibility in high-dimensional motion planning, leveraging GPU-accelerated simplicial reconstruction.

Projects

Crazy Taxi - AI Multimodal Claims Assessment | *PyTorch, YOLO, Grad-CAM, NLP* | GitHub

- Developed an AI-powered multimodal system to evaluate insurance claims, processing unstructured data from text and images to estimate damage severity and cost bands using Python.
- Implemented Deep Learning networks (YOLO, Grad-CAM) and TF-IDF to analyze consistency between visual regions and user text, detecting fraudulent images to output a final claim confidence score.

Quantum Battleship | *Qiskit, Python, Quantum Algorithms* | GitHub

- Developed a quantum algorithm applying the Elitzur-Vaidman interaction-free measurement to solve a Battleship grid optimally, demonstrating a strong theoretical foundation in quantum computing.
- Built and simulated quantum circuits using Qiskit and Python in Jupyter Notebooks to measure probe superposition and entanglement, drastically reducing classical measurements from $O(N^2)$ to $O(N)$.

NaviSight – Edge AI Wearable Device | *Python, PyTorch, YOLOv11, Audio DSP* | Website

- Fine-tuned a YOLOv11 model for real-world object detection, building an offline inference pipeline optimized for latency and resource-constrained edge devices.
- Engineered a spatial alignment feature utilizing binaural audio feedback, applying pitch and frequency variations to dynamically estimate object distance, direction, and center-alignment.

Real-Time Chat Application | *React.js, Node.js, MongoDB, Socket.IO* | GitHub

- Designed a full-stack real-time messaging application with secure JWT authentication, REST APIs, and Socket.IO for low-latency distributed communication.

Technical Skills

Languages: Python (Pandas, NumPy), C/C++, SQL, Shell, Java
AI, ML & Quantum: PyTorch, TensorFlow, Qiskit, Deep Learning, Generative AI, Large Language Models (LLMs), NLP
Developer Tools: Git, VS Code, Jupyter Notebooks, Linux, GitHub, Watsonx.ai, ROS
Core CS & Math: Data Structures, Algorithms, Linear Algebra, Probability, Statistics, Problem Solving

Achievements/Extracurriculars

- Achieved a peak rating of **1269** on **Codeforces**, solving 200+ algorithms and data structures problems.
- Secured **3rd Place** in **Megathon 2025**, building a Multi-Modal AI Solution for Estimating Damage Claims.
- Secured **AIR 4391** in JEE Mains & **AIR 10931** in JEE Advanced; Cleared **KVPY 2022** & **Pre-RMO 2019**.

Relevant Coursework

Data Structures and Algorithms | Statistical Methods in AI | Probability & Random Processes | Signal Processing | Computer Vision | Database Systems | Operating Systems and Networks | Intro to Processor Architecture | Robotics: Planning and Navigation | Data Analytics I | Object-Oriented Programming | Communication Theory